

Homework #6: Tight binding model

Due date: Nov 9th 2025

Tight bindings in the BCC and FCC lattices

1. We consider identical atoms located at points of a lattice that are body-centered cubic (**BCC: i**) and face-centered cubic (**FCC: ii**).

(a) For each case, determine the dispersion relation for s -electrons, having the general form:

$$E = -\alpha - \gamma \sum_s e^{-i\mathbf{k} \cdot \mathbf{R}_s}$$

where α and γ are the positive energies that can be calculated and $\mathbf{R}_s$ represents the vectors that connect an atom located at the origin to each of its nearest neighbors.

- (b) Find the expressions for E at the points located at the intersection of the first Brillouin zone with the directions $[100]$, $[110]$, and $[111]$ and thus the value $E(\Gamma)$ to the center of this zone.
- (c) Deduce the total width of the corresponding band, ΔE , and find the expression for the effective mass m^* in the vicinity of Γ as well as the shape of constant energy surfaces around this point.